A Comparison of Neural Architectures for Movie Poster Genre Classification *

Samson Lam

May 2024

Abstract

This study examines whether the genre of a film can be recovered from its poster alone, and at what computational cost. A convolutional network trained from scratch, four ImageNet backbones and YOLOv8 adapted by transfer learning, and the vision-language model CLIP applied zero-shot are compared with two backbones built for efficient inference, EfficientNetV2 and FastViT. The VGG16 classifier draws on the whole poster rather than its center, tolerates rotation and is largely indifferent to color. Poster design changes measurably across decades, and supplying the decade as an indicator raises the accuracy of the convolutional network by about 4 %. ResNets attain the highest accuracy, 54.5 % on the 19-genre task and 69.7 % on the simplified 3-genre task.[1]

1 Introduction

A movie poster carries a deliberate visual argument. It is composed to tell a viewer, in the moment before any synopsis is read, what kind of film this is. Genre is the part of that argument the designer most wants to land, and it is encoded in choices a human reads without effort: the palette, the typography, the arrangement of faces and the presence or absence of a horizon. This study examines whether those choices survive as a signal that a neural network can read from the poster alone.

We treat genre recognition as image classification on the poster alone. The dataset is the Letterboxd collection published on Kaggle (Garanin, 2024), from which 548,835 posters survive data cleaning, each labeled with 1 to 4 of 19 genres. We compare neural network architectures, beginning with a convolutional neural network (CNN). VGG16 (Simonyan and Zisserman, 2015), AlexNet (Krizhevsky et al., 2012) and two depths of residual network (ResNet) (He et al., 2016) supply the transfer learning. YOLOv8 (Jocher et al., 2023), the eighth release of the You Only Look Once (YOLO) detector, contributes a detection backbone adapted to classification. CLIP (Radford et al., 2021) is evaluated zero-shot, without any training on posters, and serves as the reference a fine-tuned model has to beat. EfficientNetV2 (Tan and Le, 2021) and FastViT (Vasu et al., 2023) are backbones designed for cheap inference, and they test whether the accuracy available at this task requires the parameter counts the older backbones carry. ResNet-50 attains the highest accuracy, 54.5 % on 19 genres, the two ResNets reach 69.7 % on 3 genres, and zero-shot CLIP matches the fine-tuned efficient backbones on 3 genres.

*This is a class project for the Chinese University of Hong Kong's STAT 4012 course, completed by a group of six members.

[1]Code and replication data are available at *GitHub*.

The remainder of the report is organized as follows. Section 2 describes the dataset and the data cleaning pipeline. Section 3 trains a CNN from scratch. Section 4 adapts VGG16, AlexNet, ResNet-18, ResNet-50 and YOLOv8 by transfer learning. Section 5 evaluates CLIP zero-shot and fine-tunes EfficientNetV2 and FastViT. Section 6 compares all models and concludes.

2 Data

The Letterboxd collection consists of 557,539 poster images with genre, country, language and release date tables. We keep the 552,531 posters of the dominant size $(345, 230, 3)$, 99.1 % of the data (Table 19 in the appendix). Next, movies carrying an implausible number of genres are treated as outliers. The mean genre count plus or minus 3 standard deviations spans $[-0.904, 4.006]$, so we cap the genre count at 4 per movie. This step discards 3,827 movies of the genre table, 3,696 of them among the 552,531, leaving 548,835 movies.

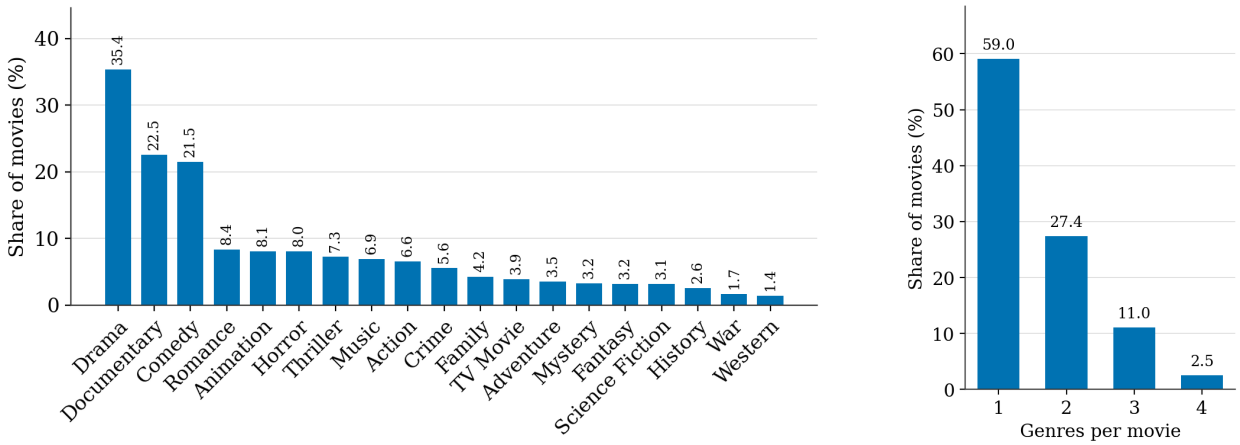


Figure 1: Distribution of genres (left) and the number of genres per movie (right).

Figure 1 displays the shares of the 19 genres on the left, with Drama, Documentary and Comedy leading, and the number of genres carried on the right, where the mass sits on one or two. Figure 2 gives the production region and the spoken language. The dataset is dominated by European and North American productions and by English-language releases. This imbalance is used in Section 4.1, where we evaluate the trained classifier separately by region and language to test its accuracy dependence.

The first release dates span from 1874 to 2031, with some posters created well in advance of the actual release date. Figure 3 illustrates their distribution by decade. On the grounds that the earlier material is sparse and stylistically distinct from the rest, the CNN of Section 3 uses only posters released after 1970 with at most 3 genres, leaving a total of 424,102 posters after all data cleaning.

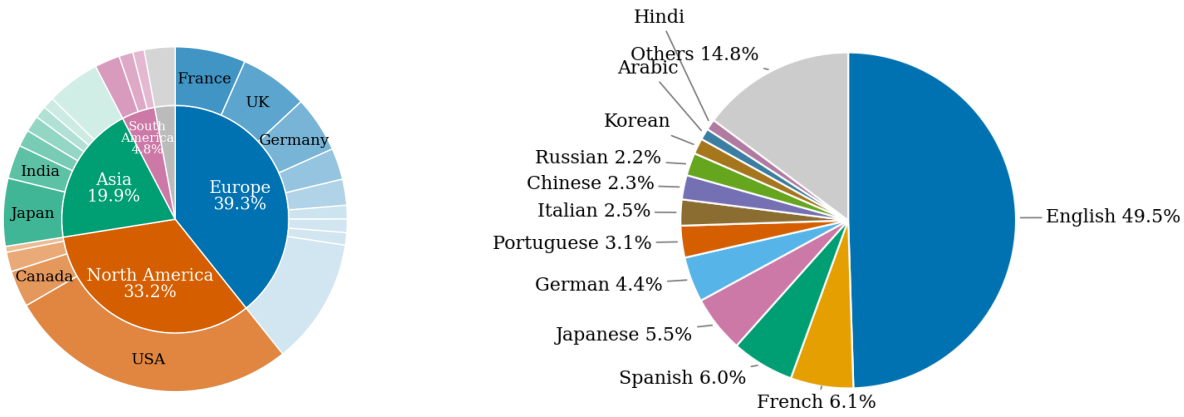


Figure 2: Distribution of regions (left) and languages (right).

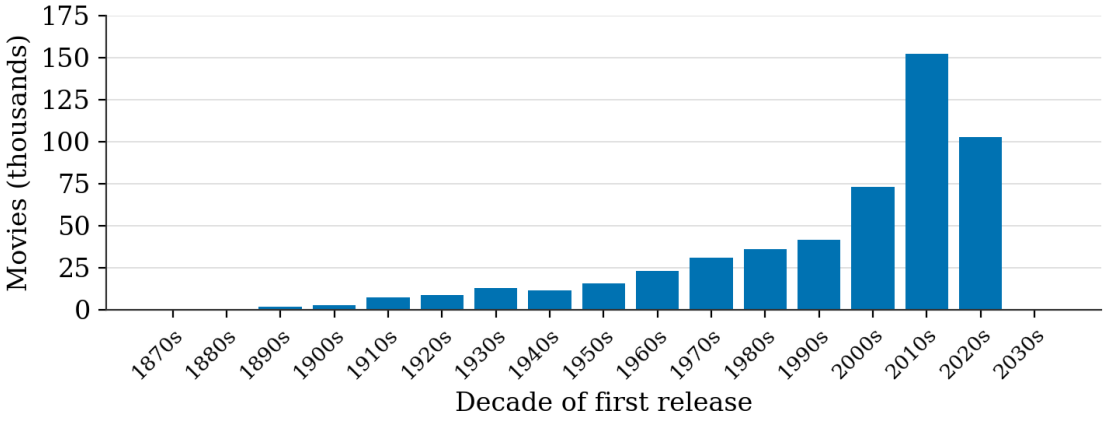


Figure 3: Movies by decade of first release.

3 Convolutional Neural Network

This section constructs a CNN from scratch, which fixes the floor every later model is measured against. Given the architecture limits, we remove two genres before training. Horror and Thriller rely on the same visual vocabulary of suspense, threat and psychological pressure and are difficult to separate even by eye, so Thriller is left out. Western is also dropped due to a lack of images.

3.1 Model

We propose a 12-layer CNN model, with the initial 9 layers having 32 filters and the last 3 layers having 64 filters. All layers use a 3×3 kernel, rectified linear unit (ReLU) activation and padding. Group normalization is applied after each layer to improve learning, and max-pooling layers downsample the feature maps. After flattening, batch normalization is applied, and the data passes into a dense layer with 32 units and ReLU activation. Dropout regularization with a rate of 0.25 is then employed to prevent overfitting. In the next layer, batch normalization is applied again, and the data passes through another dense layer with 64 units and ReLU activation. Dropout regularization with a rate of 0.5 is applied at this stage. Finally, a dense layer with the number of genres G to classify and softmax activation is added, producing predicted probabilities for each genre. The model is trained with Adam on the categorical cross-entropy loss, with accuracy as the evaluation metric. Figure 14 in the appendix draws the stack.

The sample contains 9,000 posters: 7,200 for fitting and 1,800 for testing, balanced across genres. An initial attempt with $G = 5$ genres produced accuracy, F1 and receiver operating characteristic (ROC) curves no better than a random guess, which motivated the reduction to 3 genres reported below.

Poster design is not stationary. Figure 4 places one poster from each of four decades side by side, and the progression is visible without measurement. The 1910s poster is monochrome and relies on tonal contrast. The 1950s poster carries the vivid, broadly realistic palette Technicolor made available. By the 1990s, the colors are saturated and chosen for impact. The 2010s poster returns to restraint, using gradients and narrow tonal ranges to suggest depth. Therefore, we encode the release decade as 6 indicator variables, concatenated with the flattened features (Figure 14).

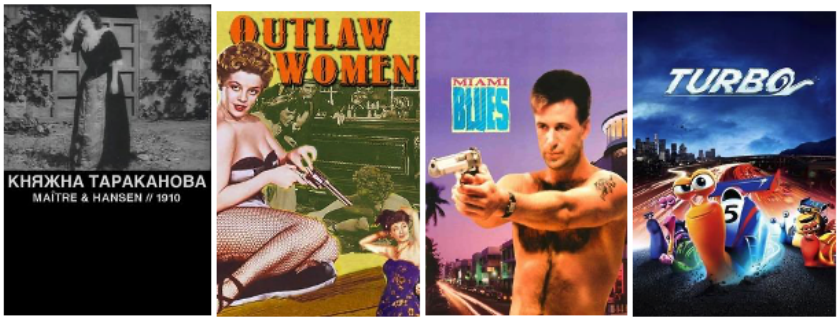


Figure 4: Poster design across four decades, from the 1910s (left) to the 2010s (right).

3.2 CNN Performance

We fit the CNN on three sets of 3 genres: Action, Drama and Romance; Comedy, Drama and Romance; and Drama, Horror and Music. Table 1 reports the area under the ROC curve (AUROC), F1, precision and accuracy for releases after 1970 and after 2000. Across repeated runs, the indicators raise accuracy by about 4 %, which we take as confirmation that release period carries information the pixels do not supply on their own. Nevertheless, the CNN remains weak, with an average accuracy of 45.2 % against 33.3 % for a random guess on 3 balanced genres. As a result, the remainder of the report turns to pre-trained models, where the representation is learned elsewhere, and only the classifier is fitted to posters.

Setting	AUROC	F1	Precision	Accuracy
Action, Drama, Romance, after 1970	60.2	42.6	42.9	44.5
Action, Drama, Romance, after 2000	66.2	46.6	45.4	44.9
Comedy, Drama, Romance, after 1970	62.0	45.6	46.3	47.3
Comedy, Drama, Romance, after 2000	62.1	45.6	45.6	46.5
Drama, Horror, Music, after 1970	62.5	45.2	45.3	46.6
Drama, Horror, Music, after 2000	53.0	41.7	42.4	41.4
Average	60.9	44.6	44.7	45.2

Table 1: Evaluation metrics of the CNN, in percent.

4 Transfer Learning

Transfer learning reuses a representation learned on a large dataset and fits only a new classifier to the current task. The pre-trained body has already encoded edges, textures and shapes, so training is shorter, and the risk of overfitting a small sample is lower.

4.1 VGG16

4.1.1 Model

VGG16 is a 16-layer convolutional network built from a single repeated element, the 3×3 convolution (Simonyan and Zisserman, 2015). Its design argument is that a stack of small kernels reaches the receptive field of a larger one while using fewer parameters and inserting more ReLU nonlinearities along the way, so depth buys expressiveness more cheaply than width. Convolutions are grouped into five blocks, each closed by a 2×2 max pool that halves the spatial extent, while the channel count doubles from block to block up to 512. Trained on ImageNet, the resulting body is a general-purpose feature extractor, and we use it here with frozen weights so only the classifier on top is fitted to posters.

We continue with 3 genres, Action, Romance and Thriller, on releases from 2000 to 2024. After flattening, the VGG16 output feeds into a dense layer with 128 units and sigmoid activation, then

dropout at 0.25, then a softmax layer with 3 outputs. Sigmoid is used in the single hidden layer because the network is shallow enough that saturation is not a concern. Table 2 gives the parameter counts.

Layer	Output shape	Parameters
Input	$(345, 230, 3)$	0
VGG16	$(10, 7, 512)$	14,714,688
Flatten	(35,840)	0
Dense, sigmoid	(128)	4,587,648
Dropout at 0.25	(128)	0
Output, softmax	(3)	387

Table 2: Layers of the VGG16 classifier with output shapes and parameter counts.

4.1.2 VGG16 Performance

Fitting minimizes categorical cross-entropy with Adam at a learning rate of 10^{-3}, in mini-batches of 32, for 50 epochs. Figure 5 plots validation accuracy and loss by epoch on the left and the ROC curves on the right, and Table 3 gives the metrics on the held-out posters. Accuracy and F1 both sit near 61 %, and AUROC reaches 79.0 %, a clear advance over the CNN in Section 3 at a fraction of the training cost.

Model	AUROC	F1	Precision	Accuracy
VGG16	79.0	61.4	62.0	61.4

Table 3: VGG16 on the held-out 3-genre sample, in percent.

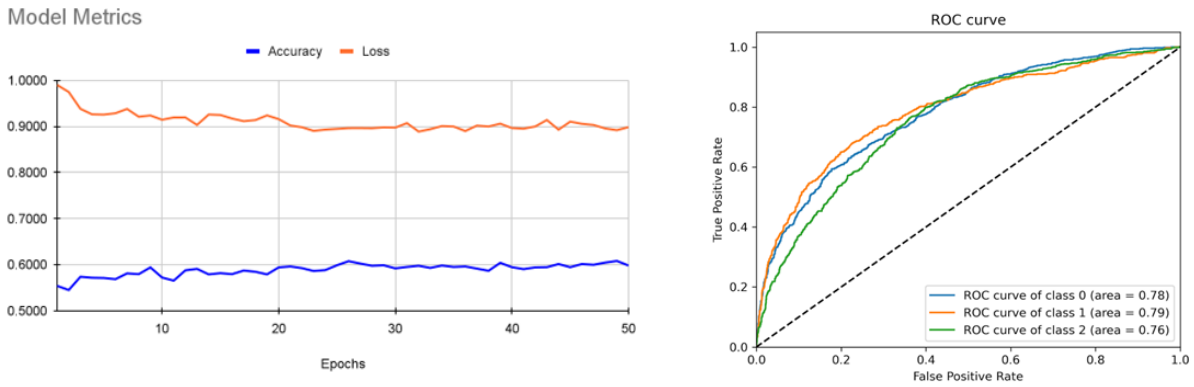


Figure 5: Fit and discrimination of the VGG16 classifier. Validation accuracy and loss by epoch on the left, ROC curves on the held-out sample on the right.

4.1.3 Model Inspection

To explain how VGG16 classifies posters, we degrade the test posters in three controlled ways and re-run the trained classifier on each, holding everything else fixed and reading the F1 score. First, we inspect the importance of color, border and orientation. Figure 6 shows one poster under each transformation beside the original.



Figure 6: A test poster in its original form and under the three transformations: monochrome conversion, blanking of 30 border pixels and rotation by 180 degrees.

Transformation	True genre	Predicted		
		Action	Romance	Thriller
Monochrome, F1 = 59.4 %	Action	482 → 409	134 → 165	184 → 226
	Romance	119 → 80	510 → 507	150 → 192
	Thriller	169 → 133	170 → 177	482 → 511
Border removed, F1 = 45.7 %	Action	482 → 202	134 → 60	184 → 538
	Romance	119 → 63	510 → 244	150 → 472
	Thriller	169 → 45	170 → 40	482 → 736
Rotated, F1 = 54.9 %	Action	482 → 526	134 → 94	184 → 180
	Romance	119 → 247	510 → 345	150 → 187
	Thriller	169 → 252	170 → 115	482 → 454

Table 4: Confusion matrices under three transformations of the test posters, written as the original count followed by the transformed count. The original F1 is 61.4 %.

Removing color barely registers (Table 4). Converting the posters to grayscale by averaging the three channels leaves the confusion matrix almost intact and F1 falls only to 59.4 %, from which we infer that the network uses color chiefly to locate edges, and that the grayscale gradient preserves those edges.

Removing the border does considerable damage. Blanking 30 pixels along each edge drives F1 down to 45.7 % and pushes most posters into the Thriller column. The classifier is therefore not

concentrating on the center of the poster in the way a human viewer does. Its filters act across the whole image and any part removed is a loss.

Turning the poster upside down costs little. F1 holds at 54.9 % and the loss is concentrated in Romance. We attribute this to Romance posters carrying human contact more often than the other two genres, and the reading of a contact between two figures is sensitive to orientation in a way that a skyline or a weapon is not.

Next, we examine whether the classifier has cultural biases, since the dataset is dominated by Western and English-language releases. Tables 5 and 6 evaluate it separately by production region and by spoken language. F1 is lowest for African and South American releases, which also have the fewest examples. Across languages, the spread is narrower, and Hindi is the only clear shortfall, while Korean releases score above the English baseline. Oversampling underrepresented regions and languages can help, but we do not attempt it here.

	Africa	Asia	Europe	North America	South America	Oceania
F1	54.7	61.3	60.3	64.8	57.6	59.2

Table 5: VGG16 F1 by production region, in percent.

	English	Spanish	Cantonese	Hindi	Korean	Thai
F1	61.5	62.1	61.4	57.0	65.8	61.6

Table 6: VGG16 F1 by spoken language, in percent.

The feature maps confirm the usual convolutional hierarchy, in which deeper layers extract more complex features (Figure 7): layer 8 responds to edges, and layer 16 responds to shapes and finer structure, isolating the face on a poster for *Ip Man*.

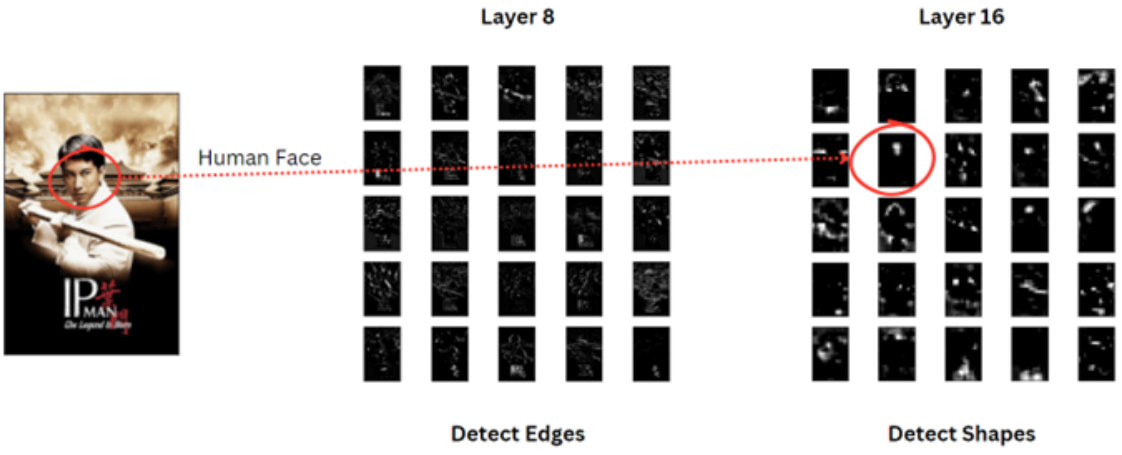


Figure 7: Feature maps from layers 8 and 16 on a poster for *Ip Man*.

Overall, VGG16 is therefore rotation tolerant, largely indifferent to color, but reliant on the

whole image rather than its center. However, it does not appear to capture the spatial relation between objects, which is where the deeper and more recent architectures in the following sections earn their place.

4.2 AlexNet and Residual Networks

While Sections 3 and 4.1 reduced the problem to $G = 3$ genres, this section also addresses the full 19-genre problem.

4.2.1 Models

AlexNet is the network that made deep convolutional models the default for image classification, winning the 2012 ImageNet competition at a top-1 accuracy of 63.3 % and by a wide margin over the runner-up (Krizhevsky et al., 2012). AlexNet's 5 convolutional layers feed 3 fully connected ones, with ReLU activations throughout and dropout in the classifier. The implementation used here sets the channels to 64, 192, 384, 256 and 256 and carries 61.1 million parameters, almost all of them in the two dense layers of width 4,096. By current standards, it is shallow and is included as the weakest of the pre-trained bodies rather than as a competitive candidate.

ResNets answer the problem that depth alone creates. Past a certain number of layers, accuracy degrades even on the training set, so the difficulty lies in optimization rather than overfitting. The remedy is to have each block learn a residual against its own input rather than a full transformation,

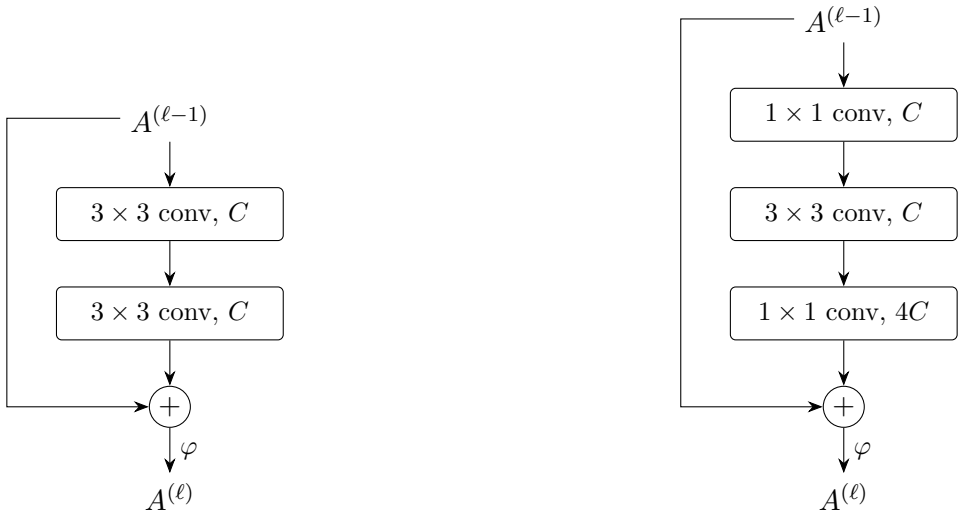
$$A^{(\ell)} = \varphi\Big(\mathcal{F}\Big(A^{(\ell-1)}; \theta^{(\ell)}\Big) + A^{(\ell-1)}\Big), \tag{1}$$

where $A^{(\ell)}$ is the output of block ℓ, $\mathcal{F}(\cdot\,; \theta^{(\ell)})$ is the stack of convolutions inside block ℓ with weights $\theta^{(\ell)}$ and φ is the ReLU, so the identity is available at no cost (He et al., 2016).

Figure 8 draws the two block types the two depths use. ResNet-18 stacks the basic block, two 3×3 convolutions of equal width. ResNet-50 stacks the bottleneck block, a 1×1 convolution that narrows the channels, a 3×3 convolution at that reduced width and a 1×1 convolution that widens by a factor of four, which buys depth at a fraction of the arithmetic. Table 7 sets out the three backbones. Figures 15 and 16 in the appendix draw the AlexNet stack and the residual stages. The pair of depths is what makes the comparison informative. If depth is what matters, the two should separate, and if the residual connection is what matters, both should stand clear of AlexNet.

	AlexNet	ResNet-18	ResNet-50
Stem	11×11 conv, stride 4	7×7 conv, stride 2	7×7 conv, stride 2
Body	five convolutions	2, 2, 2, 2 basic	3, 4, 6, 3 bottleneck
Widths	64 to 384	64 to 512	256 to 2,048
Head	two dense, 4,096	global pool, linear	global pool, linear
Parameters	61.1 M	11.7 M	25.6 M

Table 7: The three transfer learning backbones, as implemented in torchvision.



(a) Basic block, ResNet-18.

(b) Bottleneck block, ResNet-50.

Figure 8: The two residual blocks of Equation (1).

4.2.2 AlexNet and ResNet Performance

All three backbones take the poster resized to 224×224. We replace each model's softmax output with a linear layer with sigmoid activation, since a movie may carry several genres, and we use binary cross-entropy as the loss. Training runs for 5 epochs. We compare Adam and stochastic gradient descent, with the latter at a learning rate of 10^{-4}, while Adam uses 10^{-6} for AlexNet, 5×10^{-5} for ResNet-18, and 10^{-5} for ResNet-50. We track multi-label AUROC during training and validation.

Adam converges faster and to a lower loss than stochastic gradient descent for both architectures. For AlexNet, the training losses are close, but the validation metrics separate clearly. For ResNets, the gap is even wider, as shown in Figure 9. Every result reported below uses Adam.

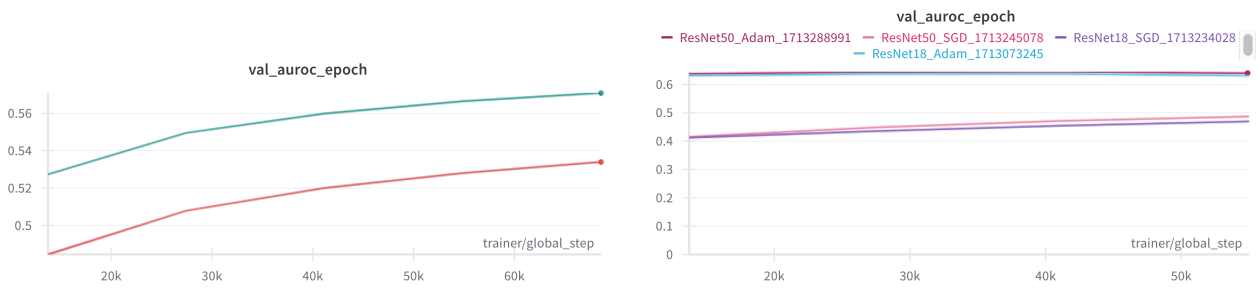


Figure 9: Validation AUROC by epoch under Adam and stochastic gradient descent, for AlexNet (left; Adam in green, stochastic gradient descent in red) and ResNet (right).

Figure 10 compares the three architectures under Adam. ResNet outperforms AlexNet throughout, consistent with greater depth allowing more abstract features and the skip connection in Equation (1) keeping the gradient usable through that depth.

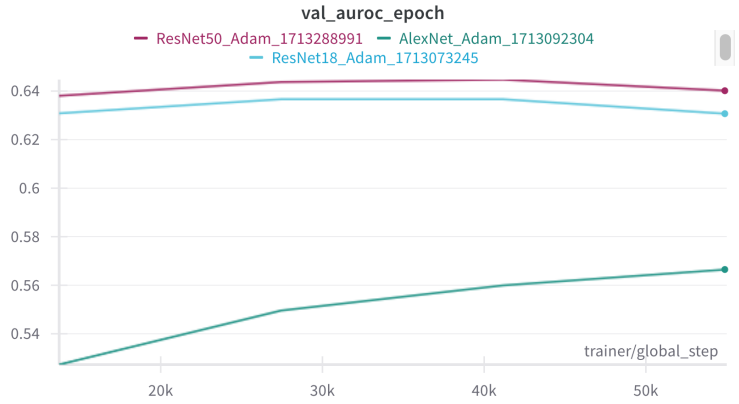


Figure 10: Validation AUROC by epoch for AlexNet and ResNets under Adam.

ResNet clearly outperforms AlexNet, but greater depth within the residual family adds little: ResNet-50 raises the 19-genre accuracy from 53.6 % to 54.5 % (Table 8) and ties ResNet-18 at 69.7 % on 3 genres (Table 9). ResNet-50 is far deeper than ResNet-18, yet their training and validation curves nearly coincide, and ResNet-50 leads by a slim margin on every genre. The gap between AlexNet and ResNet is therefore attributable to the residual connection rather than to depth alone. Overall, ResNet-18 performs almost as well as ResNet-50 with fewer than half the parameters, 11.7 M against 25.6 M (Table 7). Table 20 in the appendix gives AUROC by genre.

Model	AUROC	F1	Precision	Accuracy
AlexNet	54.5	13.4	22.7	42.9
ResNet-18	61.6	28.1	34.0	53.6
ResNet-50	62.6	29.9	33.6	54.5

Table 8: AlexNet and ResNet on 19-genre single-label classification, in percent. AUROC is computed from the hard predictions.

Model	AUROC	F1	Precision	Accuracy
AlexNet	71.6	61.5	62.8	61.8
ResNet-18	77.4	69.8	69.7	69.7
ResNet-50	77.5	69.7	70.2	69.7

Table 9: AlexNet and ResNet on single-label classification over Action, Romance and Thriller, in percent.

4.3 YOLOv8

4.3.1 Model

YOLO is a family of real-time object detectors for images and video, built for detection rather than classification. It divides the image into a grid and predicts bounding boxes with class probabilities for each cell in a single forward pass, which makes it fast enough for video. The eighth release ships a classification branch, YOLOv8n-cls, in which the detection head is replaced by a classifier over the whole image (Jocher et al., 2023). Its backbone alternates five stride-two convolutions with four C2f blocks, each of which splits its input, passes one part through a chain of bottleneck units and concatenates the results, and the head is a global pool followed by a single linear layer. The nano variant used here is the smallest of five sizes, at 2.7 million parameters against the 61.1 million of AlexNet. It is included to test whether a backbone shaped by a detection objective, which must localize objects rather than summarize a scene, reads a poster differently from a classification backbone.

4.3.2 YOLOv8 Performance

The input convention is a directory tree, with posters filed under a folder named for their label and split into training and validation folders. We implement this using the pre-trained YOLOv8n-cls model in two ways. Under the multi-label approach, a poster is copied into one folder per genre it carries, and each copy is a single-label example under the cross-entropy loss, unlike the sigmoid outputs of Section 4.2. Under the multi-class approach, it appears once, under a single genre. Both train for 10 epochs with Adam and the default cross-entropy loss.

The multi-label approach barely learns. Over 10 epochs, the training loss falls by 0.2, and top-1 accuracy rises by 4 %, settling near 35 %, and the normalized confusion matrix in Figure 11 shows that most posters are predicted as Drama, with only Animation, Comedy, Documentary, Horror and Western partly separated. Filing one poster under several conflicting labels appears to be the cause. The multi-class approach trains more cleanly, as Figure 12 shows.

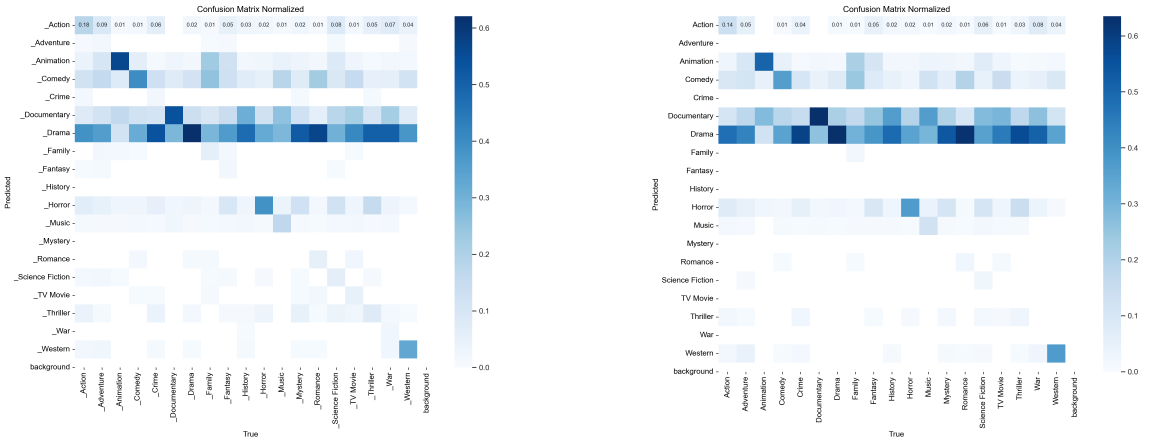


Figure 11: Normalized confusion matrices, multi-label (left) and multi-class (right).

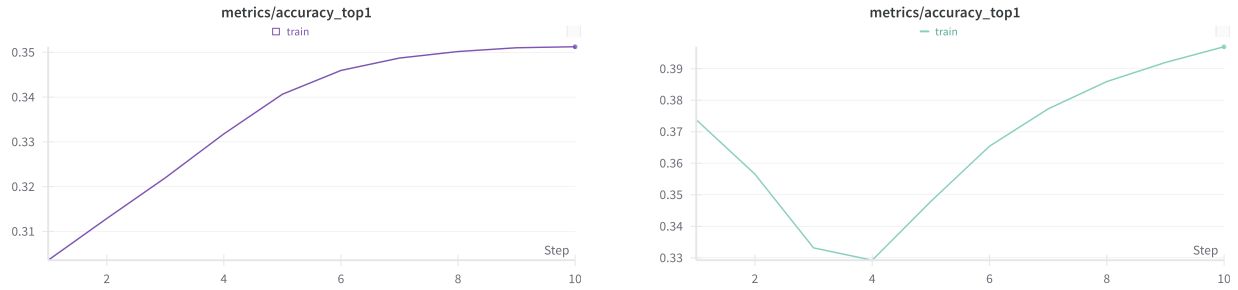


Figure 12: Top-1 accuracy by epoch, multi-label (left) and multi-class (right).

Nevertheless, Table 10 shows the multi-label model ahead on 19 genres and level or ahead on 3 genres, despite its weaker training curve. A plausible reading is that the multi-class model learns one class deeply while the multi-label model acquires broader but shallower knowledge of the label space.

Task	Model	AUROC	F1	Precision	Accuracy
19 genres	YOLOv8, multi-label	82.8	29.7	29.7	29.7
	YOLOv8, multi-class	82.0	29.0	29.0	29.0
3 genres	YOLOv8, multi-label	60.4	39.6	39.6	39.6
	YOLOv8, multi-class	53.8	39.6	39.6	39.6

Table 10: The two YOLOv8 models on single-label classification over 19 genres and over Action, Romance and Thriller, in percent. F1 and precision are micro-averaged and therefore equal accuracy.

5 Zero-Shot Prediction and Efficient Backbones

5.1 CLIP

CLIP is a multi-modal model trained to align images and text in a shared embedding space (Radford et al., 2021). Write f_I for the image encoder, f_T for the text encoder and

$$s_{ij} = \frac{f_I(X_i)^\top f_T(T_j)}{\|f_I(X_i)\| \, \|f_T(T_j)\|} \tag{2}$$

for the cosine similarity between the ith image X_i and the jth caption T_j of a batch of N matched pairs. Pre-training is contrastive. It maximizes the similarity of the N matched pairs while minimizing it across the $N^2 - N$ mismatched ones, by minimizing

$$\mathcal{L}_{\text{CLIP}} = -\frac{1}{2N} \sum_{i=1}^{N} \left[\log \frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^{N} \exp(s_{ij}/\tau)} + \log \frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^{N} \exp(s_{ji}/\tau)} \right], \tag{3}$$

with $\tau > 0$ a learned temperature. Because the two encoders share one embedding space, a classification task can be posed without any further training by writing each label as a sentence T_g and returning

$$\hat{y}(X) = \underset{g \in \{1,\dots,G\}}{\arg\max} \frac{f_I(X)^\top f_T(T_g)}{\|f_I(X)\| \, \|f_T(T_g)\|}, \tag{4}$$

as Figure 13 sets out. Figure 17 in the appendix places that step inside the contrastive pre-training of Equation (3).

That property makes CLIP the natural reference point for this study. It has never seen the poster dataset, so whatever it achieves is the accuracy available with no task-specific training, and a fine-tuned model has to beat it to justify its cost.

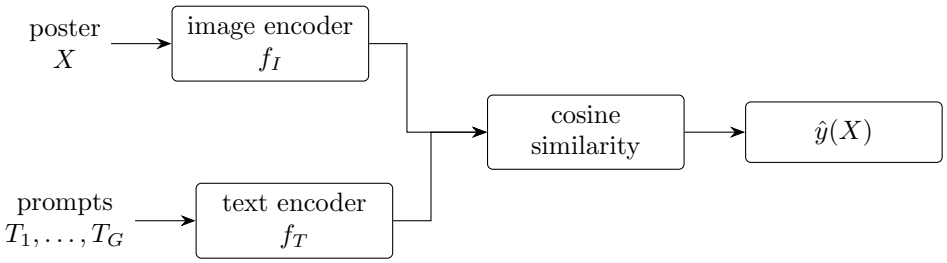


Figure 13: Zero-shot classification with CLIP. Each genre is written as one sentence, both sides are embedded, and Equation (4) returns the genre whose sentence lies closest to the poster.

The label set is written as the prompt *A movie poster of* {} *genre*, one sentence per genre, which is tokenized and passed through the text encoder. The poster is passed through the image encoder. The model returns a probability across the prompts, so an output of $[0.3, 0.3, 0.4]$ over Comedy, Documentary and Drama reads as a 0.4 probability of Drama. Four image encoders are evaluated, namely ResNet-50, ResNet-50x4 at roughly four times the arithmetic, and the base vision transformer at patch sizes 32 and 16, ViT-B/32 and ViT-B/16, which stacks 12 transformer blocks of width 768 with 12 attention heads over the flattened patches (Dosovitskiy et al., 2021), drawn in Figure 18 of the appendix. Each image encoder is paired with the text encoder pre-trained jointly with it, and the prompt is held fixed across the four.

5.2 EfficientNetV2

EfficientNet scales a network along depth, width and input resolution together rather than one at a time (Tan and Le, 2019). Depth adds layers and so enlarges the receptive field, width adds feature maps, and resolution retains more of the image. The three are tied to a single compound coefficient ϕ,

$$d = \alpha^\phi, \qquad w = \beta^\phi, \qquad r = \gamma^\phi, \qquad \text{subject to} \quad \alpha\beta^2\gamma^2 \approx 2, \quad \alpha, \beta, \gamma \geq 1, \tag{5}$$

where d, w and r are the depth, width and resolution multipliers. The constraint is what makes the scaling economical, since the arithmetic grows linearly in depth but quadratically in width and

in resolution, so a unit increase in ϕ raises the floating-point operations by a factor of about two whichever way the budget is spent.

The second version adds progressive learning (Tan and Le, 2021), which begins training at a small image size under weak regularization and raises both as training proceeds, so that the expensive large-image passes occur only once the model is ready for them. Its body mixes two block types. The Fused-MBConv block, which merges the expansion and the depthwise convolution into one ordinary convolution, is used in the early stages where the feature maps are large, and the MBConv block is used later. Table 11 gives the stage structure.

5.3 FastViT

Self-attention makes vision transformers slow and memory-intensive at high resolution. Splitting an image of side H into patches of side p yields a sequence of $n = (H/p)^2$ tokens, and self-attention costs $O(n^2)$ in that length, so halving the patch size multiplies the attention cost by a factor of 16. FastViT is a hybrid design that keeps convolutions wherever a transformer would be expensive (Vasu et al., 2023). It is staged like a residual network, halving resolution and doubling channels from one stage to the next, it replaces patch embedding with a convolutional stem, and it mixes tokens with convolutions in place of self-attention. That substitution rests on the MetaFormer observation that the transformer block keeps working when self-attention is replaced by another token mixer, a spatial multi-layer perceptron or even plain pooling (Yu et al., 2022), which Figure 19 in the appendix draws. Its own token mixer, RepMixer, uses a 3×3 depthwise convolution with batch normalization, reparameterized into a single convolution once training is complete. The variant used here is T12, with 12 blocks, the second smallest of the family after T8, in which every stage mixes tokens this way and no attention layer remains.

	EfficientNetV2-S	FastViT-T12
Stem	3×3 conv, 24 channels	3×3 convolutions, stride 4
Stage widths	24, 48, 64, 128, 160, 256	64, 128, 256, 512
Blocks per stage	2, 4, 4, 6, 9, 15	2, 2, 6, 2
Block type	Fused-MBConv and MBConv	RepMixer
Head	1×1 conv to 1,280, linear	1×1 conv to 1,024, linear
Parameters	21.5 M	7.6 M

Table 11: The two fine-tuned backbones of this section, as implemented in torchvision and in timm.

5.4 Fine-Tuning with Time Embedding

Both models are fine-tuned in the same way. We load the pre-trained weights, freeze the body so only the head is estimated, and replace the head that outputs 1,000 ImageNet classes with one that outputs the G genres. Write $\mathbf{z} = \mathcal{A}(f_{\text{body}}(X)) \in \mathbb{R}^q$ for the feature vector, where f_{body} is the frozen body and $\mathcal{A}$ is global average pooling, with $q = 1{,}280$ for EfficientNetV2 and $q = 1{,}024$ for

FastViT. For FastViT, the head is a single affine map to G outputs. For EfficientNetV2 it is two affine maps separated by dropout,

$$\boldsymbol{\eta} = U_2 \operatorname{Drop}_{0.2}(U_1 \operatorname{Drop}_{0.2}(\mathbf{z}) + \mathbf{c}_1) + \mathbf{c}_2, \qquad \Pr(y = g \mid X) = \frac{\exp(\eta_g)}{\sum_{g'=1}^{G} \exp(\eta_{g'})}, \tag{6}$$

with $U_1 \in \mathbb{R}^{1{,}000 \times q}$. Training minimizes cross-entropy using Adam for either 10 or 22 epochs.

The time embedding variant adds the release year. The year enters as a one-hot vector $\boldsymbol{\iota}(t) \in \{0,1\}^K$ over the K distinct release years of the dataset, each coordinate of which indexes a learned four-dimensional embedding, so the year contributes a vector $\mathbf{u}(t) \in \mathbb{R}^{4K}$. It is concatenated with the pooled features, so that the head of Equation (6) receives $\left(\mathbf{z}^\top, \mathbf{u}(t)^\top\right)^\top$ in place of $\mathbf{z}$ and U_1 widens to $q + 4K$ columns.

5.5 Performance

The fine-tuned models are evaluated on the posters they were fine-tuned on, so their accuracies are in-sample. Two label sets are reported: the full 19 genres and a reduced set of 3 genres, Comedy, Documentary and Drama. Narrowing the release window raises the zero-shot accuracy of CLIP slightly, from 25.5 % to 26.4 % with ResNet-50 and from 31.7 % to 34.4 % with ViT-B/32 (Table 12), consistent with a small time-varying element.

Image encoder	All periods	After 1970	After 2000
ResNet-50	25.5	25.7	26.4
ViT-B/32	31.7	33.6	34.4

Table 12: Zero-shot accuracy of CLIP over 19 genres by release period, in percent.

Over all release periods, ViT-B/16 leads on every metric on both label sets (Tables 13 and 14). On 19 genres, both vision transformers exceed both ResNets in accuracy, whereas on 3 genres ResNet-50x4 ranks last on every metric.

Image encoder	AUROC	F1	Precision	Accuracy
ResNet-50	75.9	16.7	19.3	25.5
ResNet-50x4	78.2	18.4	20.4	25.5
ViT-B/32	79.3	19.0	20.4	31.7
ViT-B/16	81.3	21.0	22.0	32.9

Table 13: Zero-shot CLIP over 19 genres, in percent.

The time embedding hurts rather than helps EfficientNetV2 at both epoch counts (Table 15), so we did not carry it over to FastViT. On 19 genres, EfficientNetV2 leads FastViT, plausibly because the 12-block FastViT is a small variant of its family, and both lead the zero-shot reference. On 3 genres, zero-shot CLIP matches both, at 60.9 % against 59.2 % for EfficientNetV2 and 60.8 % for

Image encoder	AUROC	F1	Precision	Accuracy
ResNet-50	74.4	52.8	55.8	53.2
ResNet-50x4	71.6	45.4	51.2	51.0
ViT-B/32	74.3	51.3	56.3	55.1
ViT-B/16	79.8	60.0	60.5	60.9

Table 14: Zero-shot CLIP over Comedy, Documentary and Drama, in percent.

FastViT, in 20 minutes against 135 and 100 minutes (Table 16), which suggests that on an easy label set the textual description of the genre is worth as much as task-specific fitting.

Model	Epochs	Without time embedding	With time embedding
EfficientNetV2	10	45.6	44.1
EfficientNetV2	22	46.2	30.4
FastViT-T12	10	36.9	–
FastViT-T12	22	39.8	–
CLIP, zero-shot	–	32.9	–

Table 15: Accuracy after fine-tuning over 19 genres, against the best zero-shot encoder, in percent.

Model	Epochs	Accuracy	Training and inference time (minutes)
EfficientNetV2	22	59.2	135
FastViT-T12	22	60.8	100
CLIP, zero-shot	–	60.9	20

Table 16: Accuracy after fine-tuning over Comedy, Documentary and Drama, against the best zero-shot encoder, in percent.

6 Conclusion

This study examines whether the genre of a film can be recovered from its poster alone, comparing a CNN trained from scratch, five pre-trained backbones adapted by transfer learning, zero-shot CLIP and two fine-tuned efficient backbones (Tables 17, 18, 15 and 16). A movie poster carries enough genre signal to be worth modeling but not enough to assign a reliable label. Training a CNN from scratch is not worth the compute, since VGG16, AlexNet and the ResNets, adapted by transfer learning, outperform it on every metric of Table 17. Decade indicators raise the accuracy of the CNN by about 4 %, but the time embedding lowers the accuracy of EfficientNetV2, so the drift in poster design is a small time-varying element. The model inspection showed that VGG16 is rotation tolerant, largely indifferent to color, but reliant on the whole image rather than its center, which is a different reading strategy from the one a human viewer brings to the same picture. ResNets attain the highest accuracy, 54.5 % on 19 genres and 69.7 % on 3 genres, while YOLOv8

and CLIP with ViT-B/16 lead on AUROC on 19 and 3 genres, respectively.

Two limitations qualify these comparisons. The accuracies of the fine-tuned models in Section 5 are in-sample, and the 3-genre rows of Table 17 use different genre sets. Oversampling the regions and languages on which VGG16 scores lowest is left for future research.

Model	AUROC	F1	Precision	Accuracy
CNN, average	60.9	44.6	44.7	45.2
VGG16	79.0	61.4	62.0	61.4
AlexNet	71.6	61.5	62.8	61.8
ResNet-18	77.4	**69.8**	69.7	**69.7**
ResNet-50	77.5	69.7	**70.2**	**69.7**
YOLOv8, multi-label	60.4	39.6	39.6	39.6
YOLOv8, multi-class	53.8	39.6	39.6	39.6
CLIP, ResNet-50	74.4	52.8	55.8	53.2
CLIP, ResNet-50x4	71.6	45.4	51.2	51.0
CLIP, ViT-B/32	74.3	51.3	56.3	55.1
CLIP, ViT-B/16	**79.8**	60.0	60.5	60.9

Table 17: Evaluation metrics on 3-genre single-label classification, in percent. The CNN row averages the settings of Table 1, VGG16 to YOLOv8 use Action, Romance and Thriller, and CLIP uses Comedy, Documentary and Drama. F1 and precision of YOLOv8 are micro-averaged. Bold marks the best value in each column.

Model	AUROC	F1	Precision	Accuracy
AlexNet	54.5	13.4	22.7	42.9
ResNet-18	61.6	28.1	**34.0**	53.6
ResNet-50	62.6	**29.9**	33.6	**54.5**
YOLOv8, multi-label	**82.8**	29.7	29.7	29.7
YOLOv8, multi-class	82.0	29.0	29.0	29.0
CLIP, ResNet-50	75.9	16.7	19.3	25.5
CLIP, ResNet-50x4	78.2	18.4	20.4	25.5
CLIP, ViT-B/32	79.3	19.0	20.4	31.7
CLIP, ViT-B/16	81.3	21.0	22.0	32.9

Table 18: Evaluation metrics on 19-genre single-label classification, in percent. AUROC of AlexNet and the ResNets is computed from the hard predictions, and F1 and precision of YOLOv8 are micro-averaged. Bold marks the best value in each column.

References

A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. 9th Int. Conf. Learn. Represent. (ICLR)*, 2021.

S. Garanin, 2024, "Letterboxd (Movies Dataset)," Kaggle. [Online]. Available: https://www.kaggle.com/datasets/gsimonx37/letterboxd

K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778, doi: 10.1109/CVPR.2016.90.

G. Jocher, A. Chaurasia, and J. Qiu. Ultralytics YOLOv8. (2023). Ultralytics. [Online]. Available: https://docs.ultralytics.com/models/yolov8/

A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 25, 2012, pp. 1097–1105.

A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 8748–8763.

K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. 3rd Int. Conf. Learn. Represent. (ICLR)*, 2015.

M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, 2019, pp. 6105–6114.

M. Tan and Q. V. Le, "EfficientNetV2: Smaller models and faster training," in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 10096–10106.

P. K. A. Vasu, J. Gabriel, J. Zhu, O. Tuzel, and A. Ranjan, "FastViT: A fast hybrid vision transformer using structural reparameterization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 5762–5772, doi: 10.1109/ICCV51070.2023.00532.

W. Yu et al., "MetaFormer is actually what you need for vision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 10809–10819, doi: 10.1109/CVPR52688.2022.01055.

A Appendix

A.1 Poster Dimensions

Table 19 gives the distribution of image dimensions over the 557,540 poster files of the raw download, one of which cannot be read, from which Section 2 keeps only the dominant shape.

Image size	Count	Share (%)
$(345, 230, 3)$	552,531	99.1
$(345, 230)$	4,203	0.8
$(344, 230, 3)$	752	0.1
$(344, 230)$	32	0.0
Other sizes	21	0.0
Invalid format	1	0.0

Table 19: Poster dimensions with counts and shares of the corpus.

A.2 Model Architectures

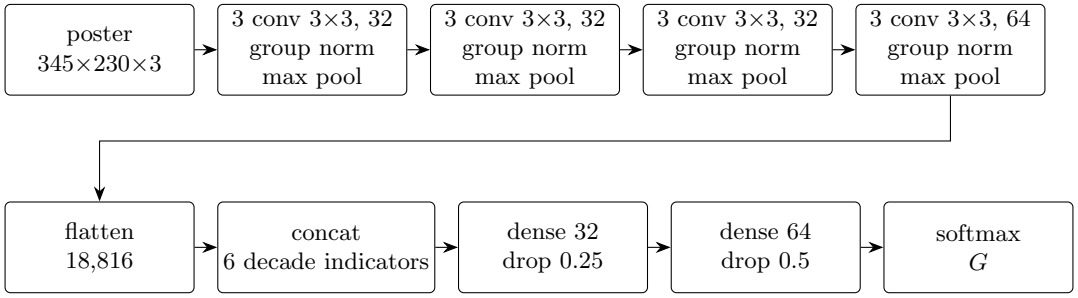


Figure 14: The CNN of Section 3.

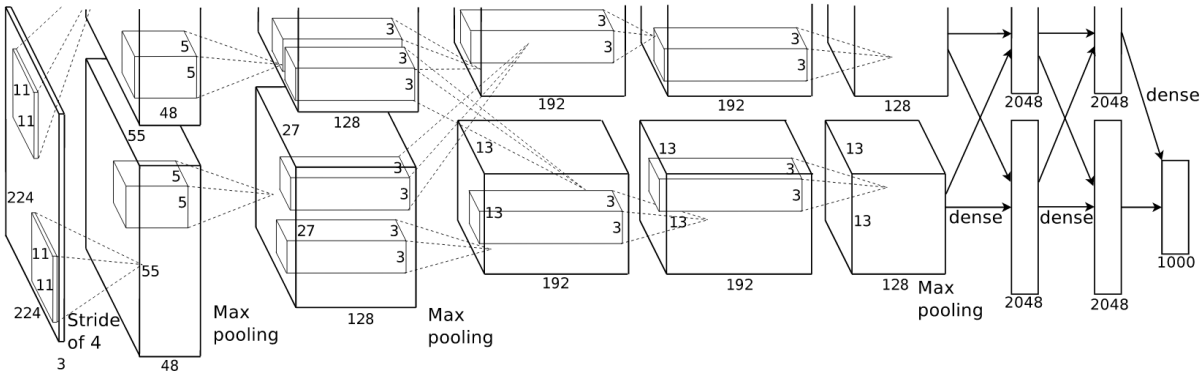


Figure 15: AlexNet, reproduced from Krizhevsky et al. (2012).

layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
conv2_x	56×56	3×3 max pool, stride 2				
		$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$
conv3_x	28×28	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times8$
conv4_x	14×14	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix}\times6$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times6$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times23$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times36$
conv5_x	7×7	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9

Figure 16: The residual architectures, reproduced from He et al. (2016).

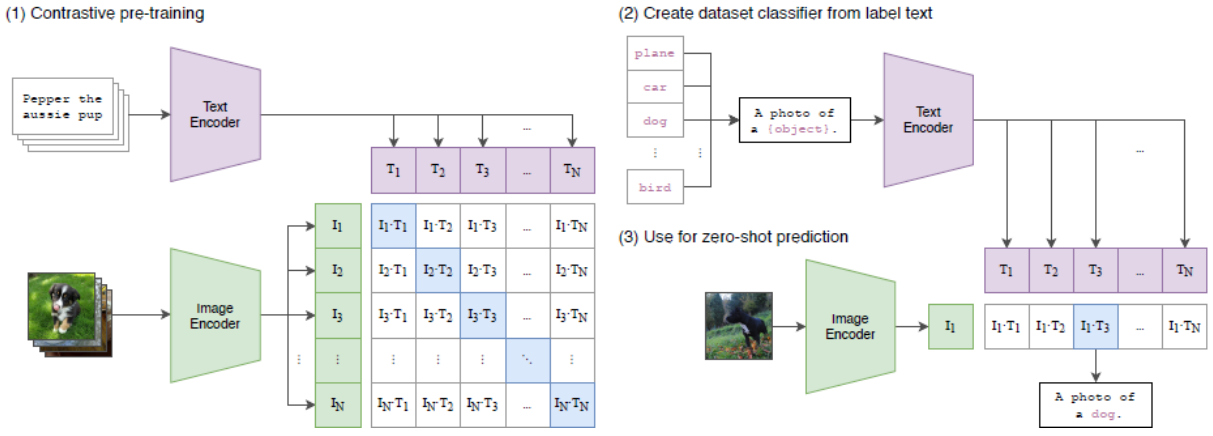


Figure 17: CLIP, reproduced from Radford et al. (2021). Panel one is the contrastive pre-training of Equation (3) and panel three is the zero-shot rule of Equation (4).

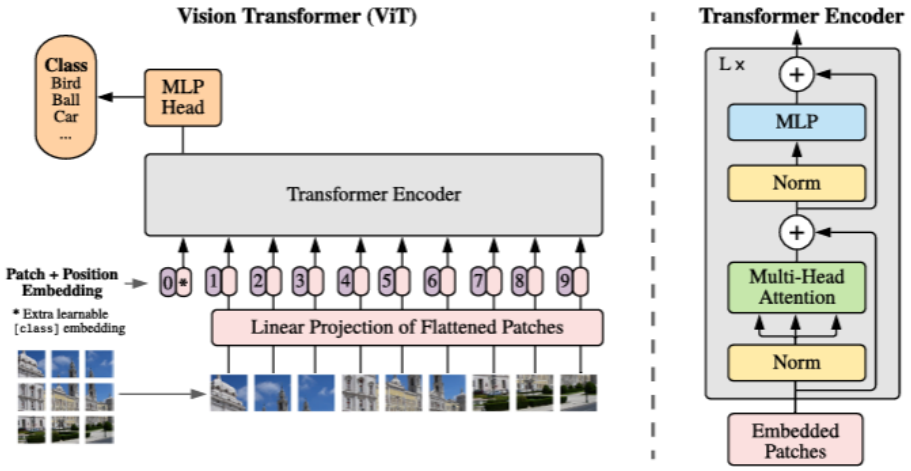


Figure 18: The vision transformer, reproduced from Dosovitskiy et al. (2021).

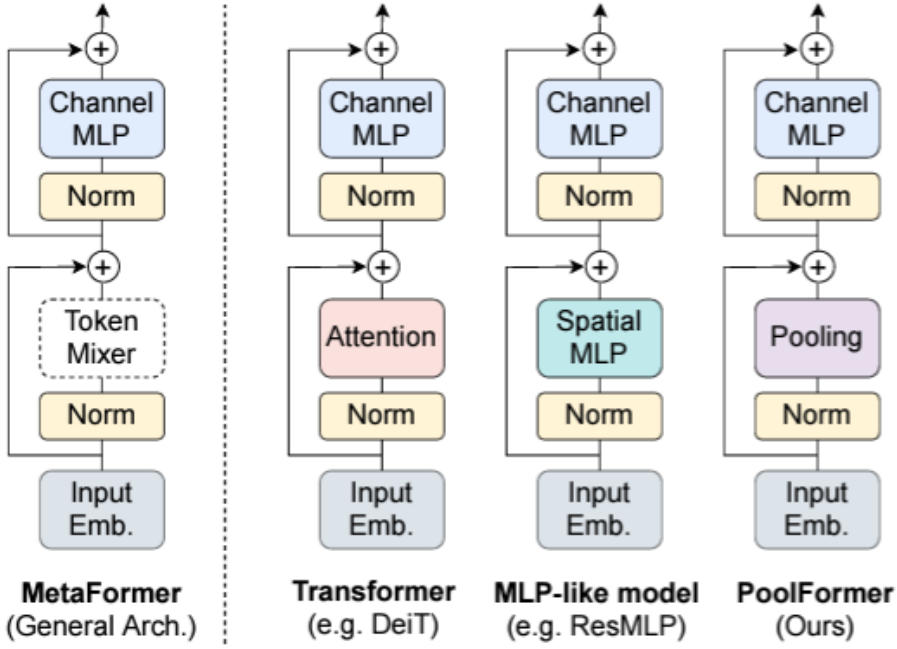


Figure 19: The MetaFormer block, reproduced from Yu et al. (2022).

A.3 AUROC by Genre

Genre	AlexNet	ResNet-18	ResNet-50
Action	77.3	83.2	84.3
Adventure	72.1	77.1	78.6
Animation	87.9	94.5	95.2
Comedy	72.6	78.0	78.4
Crime	69.1	75.7	77.8
Documentary	75.9	83.4	83.7
Drama	66.6	74.6	75.0
Family	75.6	80.3	82.3
Fantasy	66.4	72.3	74.9
History	65.2	74.8	76.8
Horror	81.6	87.0	88.0
Music	67.6	78.2	80.0
Mystery	66.0	72.4	73.9
Romance	72.6	78.9	80.0
Science Fiction	72.5	78.6	81.0
TV Movie	70.0	78.2	80.1
Thriller	74.9	80.2	81.4
War	70.9	79.7	82.3
Western	83.8	92.1	93.3

Table 20: AUROC by genre on the raw scores of the test sample for AlexNet and ResNet, in percent. Table 8 scores the hard prediction instead, hence its lower AUROC.

A.4 Sample Posters by Decade

The pool is restricted to Hong Kong releases. The drift in poster design that Section 3 measures is visible here without measurement, from painted illustration and hand-set type in the earliest decades to photographic composition in the most recent.



Figure 20: 10 Hong Kong releases sampled from the 1950s.



Figure 21: 10 Hong Kong releases sampled from the 1960s.



Figure 22: 10 Hong Kong releases sampled from the 1970s.



Figure 23: 10 Hong Kong releases sampled from the 1980s.



Figure 24: 10 Hong Kong releases sampled from the 1990s.



Figure 25: 10 Hong Kong releases sampled from the 2000s.



Figure 26: 10 Hong Kong releases sampled from the 2010s.



Figure 27: 10 Hong Kong releases sampled from the 2020s.